《数字图像处理》实验报告五 学习资料

1. 实验名称：图像复原

2. 实验示例

```
%% 噪声模型
clc
clear

r = imnoise2('gaussian',100000,1,0,1);
bins = 100;
hist(r,bins)
title('gaussian')

r = imnoise2('uniform',100000,1,0,1);
bins = 100;
figure,hist(r,bins)
title('uniform')

r = imnoise2('salt & pepper',1000,1,0.1,0.27);
bins = 100;
figure,hist(r,bins)
title('salt & pepper')

r = imnoise2('rayleigh',100000,1,0,1);
bins = 100;
figure,hist(r,bins)
title('rayleigh')

r = imnoise2('exponential',100000,1);
bins = 100;
figure,hist(r,bins)
title('exponential')

r = imnoise2('erlang',100000,1);
bins = 100;
figure,hist(r,bins)
title('erlang')
```

```
clc
clear
f = imread('image.tif');
imshow(f)
title('原始图像')

[M,N] = size(f);
R = imnoise2('salt & pepper',M,N,0.1,0);
c = find(R == 0);
gp = f;
gp(c) = 0;
imshow(gp)
title('被概率为0.1的胡椒噪声污染的图像')

R = imnoise2('salt & pepper',M,N,0,0.1);
c = find(R == 1);
gs = f;
gs(c) = 255;
imshow(gs)
title('被概率为0.1的盐粒噪声污染的图像')

fp = spfilt(gp,'chmean',3,3,1.5);
imshow(fp)
title('逆谐波均值滤波器滤波结果')

fs = spfilt(gs,'chmean',3,3,-1.5);
imshow(fs)
title('用阶为Q=-1.5逆谐波均值滤波器滤波的结果')

fpmax = spfilt(gp,'max',3,3);
imshow(fpmax)
title('用3*3最大滤波器对[被概率为0.1的胡椒噪声污染的图像]滤波的结果')

fsmin = spfilt(gs,'min',3,3);
imshow(fsmin)
title('最小滤波器滤波的结果')

%% 模糊图像 fspecial imfilter pixeldup
clc
clear

f = imread('image.tif')

PSF = fspecial('motion',7,45);  % sum(PSF(:)) = 1
```

```
gb = imfilter(f,PSF,'circular');
imshow(gb)
title('使用 PSF = fspecial(motion,7,45) 模糊后的图像')
```

2、使用维纳滤波复原技术处理图像 test.jpg

```
Image=im2double(rgb2gray(imread('test.jpg')));
LEN=21;
THETA=11;
PSF=fspecial('motion',LEN,THETA);
Blurred=imfilter(Image,PSF,'conv','circular');
noise_mean=0;
noise_var=0.0001;
Blurrednoisyl=imnoise(Blurred,'gaussian',noise_mean,noise_var)
figure,imshow(Blurrednoisyl),title('A');
estimated_nsr=0;
resultl=deconvwnr(Blurrednoisyl,PSF,estimated_nsr);
figure,imshow(resultl),title('B');
estimated_nsr = noise_var / var(Image(:));
result2    =deconvwnr(Blurrednoisyl,PSF,estimated_nsr);
figure,imshow(result2),title('C');
```

3、使用 deconvlucy 函数迭代处理模糊图像

```
I=checkerboard(8);
PSF=fspecial('gaussian',7,10);
V=0.0001;
BlurredNoisy=imnoise(imfilter(I,PSF),'gaussian',0,V);
WT=zeros(size(I));WT(5:end-4,5:end-4)=1;
J1=deconvlucy(BlurredNoisy,PSF);
J2=deconvlucy(BlurredNoisy,PSF,20,sqrt(V));
J3=deconvlucy(BlurredNoisy,PSF,20,sqrt(V),WT);
subplot(221);
imshow(BlurredNoisy);
title('A = Blurred and Noisy');
subplot(222);
imshow(J1);
title('deconvlucy(A,PSF)');
subplot(223);
imshow(J2);
title('deconvlucy(A,PSF,NI,DP)');
subplot(224);
imshow(J3);
 title('deconvlucy(A,PSF,NI,DP,WT)');
```